

How far does Crude oil volatility transmit across Food commodity markets? A Multivariate BEKK-GARCH approach

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Abstract:

The time-varying price risk emanating in nexus between crude oil and agricultural commodity prices in India, a non-food-input biofuel producing country, is analyzed using daily data from November 9, 2010 to December 31, 2015. We apply multivariate BEKK-GARCH model to study the short and long run dynamics of the volatility in both spot and futures markets of selected commodities over the sample period. The empirical findings indicate the volatility spillover from wheat futures to crude oil futures in the short-run and from crude oil futures to futures markets of maize, soybean and wheat in the long-run. The spot market linkage of selected commodities is weaker compared to futures market wherein wheat spot volatility transmits to crude oil spot market in the longer period and no spillover between crude oil-food spot market is observed in the short run. The results reveal that cross-market volatility spillover is more evident in the futures market, while own past conditional volatility is more significant in spot markets of these commodities. We also find the risk transmission from other food commodities futures to food futures prices. Our empirical findings provide strong evidence of crude oil being the reason of volatile agricultural commodity prices even in the non-food-input biofuel producing countries. The results of this study can be used to diversify the portfolios to enhance the risk-adjusted performance by both spot and futures markets' participants.

Keywords: Volatility spillover, crude oil, agricultural commodity, food price, BEKK-GARCH model, biofuel

Introduction

Energy is a crucial input for the agriculture as it is involved in all stages of the supply chain through both input and output costs, such as cost of insecticides, fertilizers, cost of production, processing and transportation (**Tyner, 2010 and Barerra et al., 2011**). This establishes a linkage between energy and agricultural commodity markets. Hence, any proliferation in crude oil prices creates more pressure on agricultural production through the route of direct and indirect costs. Recently, biofuel has emerged to be one of the catalysts for further interconnectedness of both these sectors, since agricultural commodities such as corn, wheat, soybean, sugar, rapeseed etc. are used for biofuel production, a substitute of crude oil. Therefore, any jump in crude oil prices will shift the demand towards biofuel, indirectly affecting the agricultural commodity demand and hence prices. The use of food commodities for producing biofuels has raised the world food security debate as it will have a significant impact on prices in both the markets. The rising energy intensiveness of agriculture and burgeoning biofuel production (**Baffes, 2007 and Serra et al., 2011**) have caused the increased integration between energy and agricultural commodity markets. The proponents of finance have studied the price risk and correlation structure between the trio i.e. crude oil, biofuel and agricultural commodities in U.S. (**Zhang et al., 2009; Du and McPhail, 2012; Trujillo-Barrera et al., 2012 and Gardebroek and Hernandez, 2013**), China (**Wu and Li, 2013**), Brazil (**Serra 2011 and Serra et al., 2011**), Germany (**Cabrera and Schulz, 2016**) and world (**Mensi et al., 2014**) markets. Most of the researchers in above mentioned countries found significant energy-biofuel-agricultural commodity price nexus post financial crisis.

Looking at the recent price movements in crude oil and staple food commodity markets, we observe high volatility in global as well as in Indian market in past few years. Specifically, Indian crude oil spot prices surged by 28% from September, 2011 to September, 2012.⁽¹⁾ During the same time, spot prices of Indian wheat, soybean and maize rose by 32%, 81% and 26%⁽²⁾ respectively due to the food policy and EXIM policy changes in India and global factors such as E.U. debt crisis, drought in U.S. etc. Furthermore, in September, 2014, crude oil spot prices declined by 17% followed by a decrease in spot prices of wheat (0.28%), soybean (1.60%) and maize (13.85%) on the cues of major policy changes due to the election of the new government in India and due to the turbulence in global oil markets. This co-movement between crude oil and food commodities in a non-food-input biofuel producing country (i.e. those countries which do not use food commodity in biofuel production) like India necessitates a deeper understanding of volatility spillover between these markets. Hence, it is pertinent to model the volatility risks across selected commodity markets in an emerging economy. Moreover, the price fluctuations in futures markets were more volatile than the spot markets for the above mentioned commodities. Therefore, we test the time-varying volatility and cross-market volatility spillover effect between crude oil and selected food commodities separately for both spot and futures markets of these commodities.

⁽¹⁾ <http://www.mcxindia.com/sitepages/SpotMarketHistory.aspx?sLinkPage=Y> last accessed on April 2016

⁽²⁾ <http://www.ncdex.com/index.aspx>

The nexus between the prices of crude oil and agricultural commodities has been examined by many researchers. Most of them have addressed the price level dependencies while the literature on volatility spillover is still sparse. In this paper, we examine the cross-market volatility persistence and spillover between crude oil and food commodities in Indian commodity market using multivariate BEKK-GARCH model. This study contributes to the existing literature in following ways:

First, the current set of studies in this area is limited to food-input biofuel producing countries (i.e. countries which use food commodities as an input for biofuel). Based on the findings of these studies, an investor who possess cross-country portfolio of these commodities will not be able to have an optimal hedge ratio as these findings are applicable only in food-input biofuel producing countries. Hence, our study adds onto the literature by examining the volatility dynamics between crude oil and food commodities in a non-food-input biofuel producing country (i.e. the countries which do not use food commodities in producing biofuels).

Second, alongwith the financialization of commodity market, the interconnectedness of energy and agricultural commodity markets has strengthened (Nazlioglu et al., 2013) in the aftermath of the food crisis (2006) followed by the financial crisis (2008). Hence, we expect the volatility spillover between futures prices of selected commodities due to the increased futures trading in the selected commodities. However, most of the studies on price level dependencies and volatility dynamics between energy and agricultural commodity markets are based on the spot market. A few researchers have considered futures markets in developed countries. To the best of our knowledge, there is no study which examines the volatility dynamics between futures prices of crude oil and food commodities in the context of an emerging economy. Therefore, we contribute to the available literature by analyzing the volatility dynamics between crude oil-food commodities in futures market. The findings from this study will be useful for index investors and speculators investing in emerging markets.

Third, we investigate this relationship separately for spot and futures markets in India due to two reasons: first, the profile of the participants operating in both the markets, are different. Spot market is majorly dominated by traders and producers and a major chunk of trading in futures market is performed by speculators and index investors. Second, since commodity spot and futures markets are not integrated perfectly in India and the volatility of futures prices of selected commodities has also increased in recent period, we expect different volatility dynamics in both the markets for energy-food commodity nexus. Hence, we study the volatility spillover separately for spot and futures markets of these commodities. To the best of our knowledge, our study will be the first one to provide hedging strategies to both spot and futures market participants.

Finally, we compute hedge ratios and portfolio weights using time-varying conditional variances and co-variances obtained from multivariate BEKK-GARCH model. These can be used to create

more diversified portfolios and hedge the crude oil risk and food commodity risks more effectively in non-food-input biofuel producing countries.

The remainder of this paper is organized as follows. The next section gives a brief overview of recent literature on spillover effects between agricultural and energy markets. Section 3 describes the empirical model followed by data and descriptive in section 4. In section 5, the empirical model has been developed using real data. Section 6 concludes.

Literature Review

The nexus between energy and agricultural commodity prices has been extensively studied by the researchers using empirical methods. A major chunk of the literature has examined the price level interdependencies (Baffes, 2007; Abbott *et al.*, 2008; Harri *et al.*, 2009; Gilbert, 2010; Ciaian and Kancs, 2011; Reboredo, 2012; Hassouneh *et al.*, 2012; Kristoufek *et al.*, 2012; Pala, 2013, Wang *et al.*, 2014 and Tadesse *et al.*, 2014). There are only few researchers who have tested volatilities and their spillover effects between markets, out of which most of the studies have been done on those biofuel producing countries that use food commodities as an input for biofuels (i.e. food-input biofuel producing countries). For instance, U.S. uses corn for producing ethanol and soybean for biodiesel; Brazil uses sugar for producing ethanol; China uses wheat, corn, sweet sorghum and cassava for producing ethanol. However, the empirical analysis of these risk factors for non-food-input biofuel producing countries is sparse. In the following, a brief overview of the related strand of recent empirical framework on volatility spillover between energy and agricultural commodity market is discussed.

The behavior of crude oil prices and its volatilities should affect the movements in agricultural commodities prices significantly. The vice-versa may also be true. **Wu and Guan (2011)** examine the volatility spillover from U.S. crude oil futures prices to corn cash price and futures price using parsimonious BEKK-GARCH model and find that volatility transmission effect from crude oil futures to corn cash price and futures price are almost equal and significant. But, only a small proportion of corn price volatility gets influenced by the crude oil price volatility. Similar results are found by **Du et al. (2011)** who investigate the linkage between crude oil futures, soybean and corn futures price volatility for U.S. commodity market for the period 1998-2009. The authors used stochastic volatility models using Bayesian Markov chain Monte Carlo method to estimate this relationship. They detect spillover effect from crude oil to corn to wheat market after the fall of 2006 and support them by inter-connectedness of the markets due to increased ethanol production. **Trujillo-Barrera et al. (2012)** examine the price and volatility spillover between U.S. ethanol and corn market by supplying oil shocks as an exogenous variable in VECM-BEKK-GARCH model. They document the price transmission from crude oil to corn and ethanol and volatility spillover from corn to ethanol. As for **Serra et al. (2011)**, the authors support the unidirectional price and volatility spillover from crude oil and sugar market to ethanol market in Brazil using VECM together with BEKK-GARCH model. They also state that ethanol has strengthened the relationship between energy and food markets. **Serra (2011)**

extends the findings of Serra et al. (2011) by applying semiparametric GARCH model given by Long et al. (2011), which is more robust. Their findings are similar to Serra et al. (2011). Similar to Trujillo-Barrera et al., (2012), **Wu and Li (2013)** using BEKK-MVGARCH model reported positive volatility spillover from crude oil to corn prices in Chinese commodity market for 2003 to 2012. In order to examine the impact of negative and positive shocks in crude oil prices on agricultural commodities and to inspect the feedback of agricultural commodities to the continuous and jump behavior of crude oil returns in China, **Zhang and Qu (2015)** combine the jump model (ARJI) with the GARCH model which combines the continuous process with the jump process. Using ARJI-GARCH model, they document that crude oil shocks affect cash crops (soybean, bean pulp, cotton and natural rubber) more than food crops (wheat and corn) for the period September 23, 2004-April 3, 2014. **Chen et al. (2010)** further addressed the issue of volatility spillover between global corn, soybean, wheat and WTI crude oil futures prices. Based on the results of Autoregressive Distributed Lag (ARDL) model, they conclude that volatility transmits from crude oil and other grains to grain market. **Ji and Fan (2012)** focus on the linkage between WTI crude oil futures prices and CRB non-energy indexes during pre and post financial crisis period. Using bi-variate EGARCH model, they document significant volatility spillover from crude oil to non-energy market. They also find the weakening effect of US dollar index, an exogenous shock variable, on commodity markets in post-crisis period. **Nazlioglu et al. (2013)** discuss the connection between global oil, corn, soybean, wheat and sugar prices. The results of variance in causality test and impulse response function show the unidirectional flow of volatility from oil to agricultural commodities after 2006, the post-crisis period.

Busse et al. (2010) analyze the nexus between German energy and agricultural markets (rapeseed, rapeseed oil, soybean and soybean oil) using DCC-GARCH framework. The findings of the univariate GARCH framework show that conditional volatility of all series is the function of their own past volatility and own market shocks. The correlation between crude oil and agricultural markets has increased for the sample period. Therefore, the results support the idea of increasing integration between agricultural and energy markets. **Du and McPhail (2012)** test the association between U.S. ethanol, gasoline, corn and crude oil futures prices by applying DCC-MGARCH with structural break. The findings reveal the strengthened nexus post March 2008 for the U.S. market. **Mensi et al., (2014)** examine the return and volatility relationship between international energy and agricultural commodity spot markets using VAR-BEKK-GARCH and VAR-DCC-GARCH models for the period January 3, 2000 to January 29, 2013. They also investigate the impact of OPEC announcements on such nexus. The results show significant linkages between energy and cereal markets. Moreover, the OPEC announcements affect oil markets and the nexus between oil-cereal markets. In line with Du and McPhail (2012), **Cabrera and Schulz (2016)** studied the volatility linkage between crude oil, rapeseed and biodiesel spot prices in German market using Asymmetric dynamic conditional correlation-GARCH model and Multivariate multiplicative volatility model. They find that volatility increases in all markets during turbulent times and remains stable during tranquil periods. The

results show weak linkage of volatility of biodiesel with volatility of rapeseed and oil and, strong linkage of volatility of oil with rapeseed in recent periods.

The studies reported above found unidirectional volatility spillover from energy to agricultural commodity markets in U.S., Brazil, China, Germany and World market. In contrast to above studies, **Zhang et al. (2009)**, using a multivariate BEKK-GARCH model, investigated the price transmission and volatility spillover between weekly wholesale prices of U.S. ethanol, soybean, corn, oil and gasoline. They discover the volatility spillover from agricultural commodities to energy prices in the short run. But, they do not find any spillover from ethanol to corn and soybean.

Kaltalioglu and Soytaş (2011) studying world agricultural raw material price index, food spot price index and oil spot price index for the period 1980 to 2008 concluded no volatility spillover from oil market to food and raw material markets. The researchers used Granger causality in variance approach used by Cheung and Ng (1996) for the analysis. In a more recent time period, volatility transmission effects between the U.S. crude oil and agricultural commodity markets are analyzed by **Gardebroek and Hernandez (2013)**. The authors used T-BEKK model and Dynamic conditional correlation to examine the volatility spillover between crude oil and corn prices for the period 1997 to 2011 and found no spillover from former to latter. **Zhao and Goodwin (2011)** seek to determine the structure of volatility spillover between the implied volatilities of U.S. corn and soybean option prices using Threshold VAR model with four volatility regimes. They conclude volatility spillover from corn to soybean at lower levels and from soybean to corn at higher levels of volatility. They find mixed results for the whole sample period (May, 2001 to January, 2010) depending upon the level of volatility.

The existing literature shows the strong evidence of volatility spillover from energy market to agricultural commodity market, wherein biofuels strengthen this relationship. Some of the studies have reported mixed results. In existing studies, biofuel has been given more emphasis for energy-food nexus. However, we expect this nexus to be true in non-food input biofuel producing countries too due to increased integration of financial markets and energy-intensiveness of agricultural crops. But, this phenomenon has not been empirically examined in those countries which do not produce biofuels from food commodities. Especially for the Indian commodity market, there is no study which studies the volatility linkages between energy and agricultural commodities prices.

The Empirical Model

Multivariate Generalized Autoregressive Conditionally Heteroscedastic (MGARCH) models have extensively been used in the available literature to understand the interdependence between the volatilities of energy and agricultural commodity markets. Majorly, the researchers have employed BEKK-GARCH model proposed by Baba, Engle, Kraft and Kroner (1990) and put forward by Engle and Kroner (1995) (Zhang et al., 2009; Wu and Guan, 2011; Serra et al. 2011;

Trujillo-Barrera et al., 2012; Gardebroek and Hernandez, 2013 and Wu and Li, 2013) and DCC-GARCH model defined by Engle (2002) (Busse et al., 2010; Du and McPhail, 2012; Cabrera and Schulz, 2016) for this purpose. For our study, we use BEKK model proposed by Engle and Kroner (1995) to examine the dynamic structure of volatilities between crude oil, wheat, maize and soybean in Indian market. Compared to other models, BEKK model is appropriate to identify the volatility within and across markets as it captures own and cross-market volatility spillovers along with the persistence between markets. Moreover, it also ensures the positive definiteness of the conditional variance matrix H_t . The empirical model used in this study consists of two components. A vector autoregression (VAR) with five lags for spot and one lag for futures returns is employed to model the returns. The number of lags has been selected based on Akaike Information Criterion (AIC) and Final Prediction Error (FPE) methods. Inclusion of VAR allows for cross-correlations and autocorrelations in the returns. A multivariate GARCH model has been applied to capture the time-varying variances and co-variances. The model is represented below:

$$r_t = E(r_t/\Omega_{t-1}) + \varepsilon_t \quad (1)$$

where

$$\varepsilon_t/\Omega_{t-1} \sim N(0, H_t)$$

r_t represents a 4*1 vector of price returns for wheat, corn, soybean and crude oil, Ω_{t-1} is the set of all relevant information known at time t-1. $E(r_t/\Omega_{t-1})$ denotes a 4*1 conditional expectation vector based on information set Ω_{t-1} at time t-1. ε_t is a 4*1 vector of forecast errors for the best linear predictor of r_t , conditional on the relevant information known in the past period Ω_{t-1} . H_t is the conditional variance-covariance matrix. We apply GARCH (1,1) specification as it is the best suited. The BEKK model is represented below:

$$H_t = CC' + \sum_{j=1}^q \sum_{i=1}^k A'_{ij} \varepsilon_{t-j} \varepsilon'_{t-j} A_{ij} + \sum_{j=1}^p \sum_{i=1}^k B'_{ij} H_{t-j} B_{ij} \quad (2)$$

Where C, a 4*4 lower triangular matrix contains the constants c_{ij} . The idea behind decomposing the constant term into a product of two triangular matrices is to ensure the semi-definiteness of conditional variance-covariance matrix H_t . A is a 4*4 matrix of ARCH parameters corresponding to the elements α_{ij} and α_{ii} which represent the degree of innovation between two markets i and j and own innovations respectively. B is a 4*4 matrix of GARCH parameters comprising of the elements β_{ij} and β_{ii} that measures the volatility persistence between markets i and j and own volatility persistence respectively. We also compute hedge ratios and portfolio weights from the time-varying conditional variances obtained from the multivariate BEKK-GARCH model, which can be used by portfolio investors in designing their dynamic hedging strategies and portfolio risk minimization.

Data and Statistical Properties

Several reasons have impelled us to choose Indian commodity market to study the volatility spillover among crude oil and food commodities. First, the choice of India is accentuated by the fact that it is highly sensitive to volatility in crude oil prices due to its oil dependence. India is one of the largest importers of crude oil and imports about 70 percent of its total oil consumption. Therefore, Indian crude oil prices may be function of the global crude oil prices which further may affect Indian food prices through supply side factors. Second, an urban and rural Indian household incurs around 38.5% and 48.6% respectively of total consumption expenditure on food alone. Hence, both crude oil and food prices are very crucial to Indian households as fluctuation in these prices will have a direct impact on their level of consumption. Third, the existing studies on energy-food nexus are based on the food-input biofuel producing countries. But, we expect the presence of this nexus in non-food-input biofuel producing countries too and since India is one of the fastest growing economy, it is vital to test this relationship here. We develop separate model for spot and futures prices of selected commodities as the players in both the markets are different. The spot market is majorly dominated by traders and producers while speculators and index investors occupy major share in futures market. Hence, it is essential to examine the association between crude oil and food commodities spot and futures markets separately. Most of the researchers have investigated the linkage between crude oil, corn, wheat and soybean in food-input biofuel producing countries. Therefore, we select the same set of commodities to analyze this nexus in non-food-input biofuel producing countries.

The empirical analysis is based on daily closing spot and near-month futures prices of crude oil (CS and CF), maize (MS and MF), soybean (SS and SF) and wheat (WS and WF). All prices are expressed in Indian rupee terms. Continuous West Texas Intermediate crude oil futures contract are used to measure crude oil futures prices (Rupees per barrel). Crude oil spot and futures prices are obtained from the Multi-commodity Exchange India Ltd. (MCX). Wheat, maize and soybean prices are measured by the continuous futures contract on the WHTSMQDELI (rupees per ten metric ton), Maize-feed/industrial grade (rupees per ten metric ton) and Soybean-IDR (rupees per ten metric ton) contracts respectively. All agricultural commodities spot and futures prices are collected from National Commodity and Derivatives Exchange Limited (NCDEX). The sample period ranges from November 9, 2010 to December 31, 2015 (1070 observations). The availability of data for futures prices has restricted our sample period.

The continuously compounded daily returns have been calculated by taking the logarithmic transformation of two immediate prices. Fig. 1 shows the daily movements in the spot and futures returns of crude oil, maize, soybean and wheat over the sample period. All commodities spot and futures returns exhibit similar trends except maize, suggesting that spot and futures returns are highly correlated. In August-September 2012, we can observe sharp upward movement in food commodity returns, corresponding to the Indian EXIM policy changes, U.S. drought, increased investment in Indian commodity futures market, while synchronously crude

oil returns exhibit slight upward movement. On the other hand, the squared daily returns of the selected time series are plotted in Fig. 2, indicating the changes in volatility across time. Some large spikes in volatility are visible for crude oil spot and futures prices in the year 2014-15 as a result of fear of disruption in supply of crude oil due to unrest in Iraq.⁽³⁾ Indeed, all the series show the evidence of volatility clustering. This phenomenon is parameterized by designing asymmetric-GARCH model.

The descriptive properties of the spot and futures return behaviors are presented in Table 1. The daily mean of the selected series varies between -0.0430 and 0.0541 with maize spot returns having the highest mean. Besides, crude oil spot returns yield lowest daily average over the sample period. Crude oil spot has the maximum risk (standard deviation). With respect to food commodities, futures returns exhibit more risk compared to respective spot returns, as is evident from the standard deviation of selected food commodities. The negative value of skewness for all series, except for maize futures and wheat spot, indicates the asymmetric nature of time series. The value of kurtosis is more than 3 for all series which indicates that all series have peaked distributions and fat tails. The Jarque-Berra test statistics shows non-normality of returns of all selected commodities. To examine the non-linear chaotic dynamics of the time series, we use Brock, Dechert and Scheinkman test (BDS, 1987). The BDS test statistics for all the series fail to support the null hypothesis of independent and identical distribution (white noise process), except for soybean futures, which confirms the presence of non-linear structure in all selected series. Hence, application of a GARCH-family model is appropriate for empirically modeling the results of preliminary analysis such as volatility clustering, fat-tails and volatility persistence for spot and futures returns of crude oil, maize, soybean and wheat.

Unconditional correlation between futures returns is comparatively more than the spot returns. The correlation between crude oil and soybean futures and, between crude oil and maize futures is higher (see Table 2). Similarly, it is higher for squared daily returns (see Table 3). With respect to spot returns, the correlation between crude oil and soybean (see Table 2), between crude oil and maize and, between maize and wheat (see Table 3) is higher than the other pairs of commodities. Since financial time series are non-stationary in nature, we apply Augmented Dickey-Fuller test (ADF, 1979) and the Phillips and Perron test (PP, 1988) to examine the unit root in the series. The results suggest that all the return series are stationary (see Table 4).

Results and Discussion

This segment of the paper reports the estimation results obtained from the full multivariate BEKK-GARCH model for crude oil and food commodity returns. We also provide the hedging strategies and portfolio weights to the portfolio managers in the following section.

As mentioned earlier, spot and futures market of crude oil and food commodities are examined in this study. We developed two multivariate full BEKK-GARCH models to test the return and volatility spillovers, where one model contains the spot returns of crude oil and food

⁽³⁾ http://www.business-standard.com/article/economy-policy/oil-subsidy-to-drstically-fall-during-2014-15-114101300586_1.html

commodities and the other includes the futures returns of the selected commodities. The results of multivariate BEKK-GARCH model are reported in Table 5. The mean equation (see Panel A of Table 5) of futures prices show that the past futures price returns can not be used to predict future returns as none of the m_{ii} coefficient is significant, this result supports weak-form of efficient market hypothesis which implies that previous period rate of return does not affect current period return. On the contrary, soybean spot return and crude oil spot returns contradict the weak-form market efficiency as their one-period lagged returns (represented by m_{33} and m_{11} respectively) significantly impact the contemporaneous returns. These results suggest that past returns can be used to forecast the future returns in the spot market but not in the futures market of the selected commodities. Nevertheless, only the returns of wheat futures market are influenced by the crude oil futures market returns. The coefficient of one-period lagged returns of the crude oil futures market (0.0585) is positive and significant for wheat futures market which suggests that investors should observe the information in the crude oil futures market to determine the wheat futures returns. Moreover, in spot market, bi-directional relationship is observed between crude oil returns and soybean returns where crude oil spot returns have negative (-0.0336) but significant influence on the soybean spot returns and soybean spot returns have positive (0.1120) and significant impact on crude oil spot returns, indicating that investors in these markets should follow the news in both crude oil and soybean spot markets. In addition, the coefficient of wheat spot in soybean spot equation (0.3124) shows the positive and significant impact of previous period wheat spot returns on the immediate period soybean spot returns. Meanwhile, the futures returns of crude oil, maize and soybean are not affected by any of the cross-market futures returns. Similarly, spot returns of maize and wheat are not influenced by the cross-market spot returns. Overall, the findings reject the null hypothesis of cross-market returns spillovers in futures market (with the exception of wheat futures returns which respond to the innovations in crude oil futures market) but do not reject it in spot market as significant cross-market mean spillovers are found in the crude oil, soybean and wheat spot market.

Conditional volatility can be explained by short-term and long-term persistence. Short-run persistence is represented by the ARCH effects (α_{ii} and α_{ij}) and the long-run by GARCH effects (β_{ii} and β_{ij}). Examining the conditional variance equations (see Panel B of Table 5), futures markets of crude oil (β_{11}), wheat (β_{44}) and maize (β_{22}) show strong evidence of significant role of past own conditional variance for the respective markets, indicating long-term volatility persistence for these markets. The persistence of crude oil and wheat futures markets is very high and close to one, suggesting a long memory process and signifying that changes in current volatility has an impact on the future volatility over the long-run. More specifically, crude oil futures is the most volatile market in the long-term (0.9692) followed by wheat (0.7904) and maize (0.2060). In contrast, the coefficient of long-term volatility persistence for soybean futures (β_{33}) is insignificant, implying that the future volatility of soybean futures is not the function of its current volatility. Similar to futures markets, spot markets of the selected commodities also confirm the significant long-run persistence in crude oil, maize, soybean and wheat markets. More precisely, all selected spot markets have very high persistence which is close to one,

connoting that shocks in past volatility are reflected into current volatility. The extent of long-run persistence is very high in crude oil spot market (0.9707) followed by soybean (0.8386), wheat (0.8038) and maize (0.7810). The long-run persistence in spot and futures markets differs significantly for soybean and maize. The past volatility of soybean futures is insignificant in explaining the changes in current volatility while for soybean spot it is highly significant at 1% level of significance and the coefficient is close to one. With respect to maize, the level of significance of GARCH term is more for spot market (1%) compared to futures (5%) and the value of coefficient is low in futures (0.2060) while it is close to one for spot market (0.7810). The ARCH term (α_{ii} and α_{ij}) which represents the short-run persistence is also crucial in explaining the conditional volatility. For futures market, maize (0.3810) shows the most amount of persistence in own conditional variance and for spot market, wheat (0.7224) and maize (0.4785) show the most amount of persistence in own conditional variance followed by crude oil (.1940) and soybean (-0.1157). For crude oil futures, crude oil spot, maize futures, soybean spot, wheat futures and wheat spot markets, β_{ii} values are larger than their respective measured α_{ii} values, implying that own conditional volatility long-run persistence (GARCH terms) is higher than short-run persistence (ARCH terms). The short-run persistence is larger than the long-run persistence for maize futures and soybean futures markets. The above findings suggest that in the spot market of the selected commodities, long term persistence is more than the short term while in futures market ARCH term is more than the GARCH term for maize and soybean futures.

As tabulated in Panel B (see Table 5), volatility equation parameters α_{ij} and β_{ij} represent the cross-market volatility spillover effects among selected commodities in the short-run and long-run respectively. The findings reveal that the innovations in the past wheat futures prices have positive and significant effect on crude oil futures volatility (α_{14} : 0.1928) in the short run. Meanwhile, the volatility of soybean futures is influenced by past volatility of maize futures (α_{32} : 0.4984) and wheat futures (α_{34} : 0.1927); wheat futures by maize futures (α_{42} : -1.0369) in the short run. In the longer period, volatility spillovers from crude oil futures to maize futures (β_{21} : 0.2608), soybean futures (β_{31} : -0.2293) and wheat futures (β_{41} : 0.2042). The bidirectional spillover was observed between maize and soybean futures and, maize and wheat futures. The volatility gets transmitted from maize spot to wheat spot (α_{42} : 0.2058) and crude oil spot (β_{12} : -0.0522) in short and long-run respectively. Our findings indicate that cross-market volatility spillover is more evident in the futures market while own conditional volatility spillover is more significant in spot markets of these commodities. The univariate diagnostic test for standardized residuals reports that there is no evidence of serial correlation in the standardized residuals of all chosen commodities except for spot markets of maize, soybean and wheat (see Table 6). The serial correlation test results for standardized residuals squared shows the presence of serial correlation only for maize spot market.

The plots of time-varying conditional correlations for the crude oil and food spot and futures markets exhibit significant movements in conditional correlations over the sample period (see Fig.3). It can be noticed from the plots that volatility is bundling-up in all the series. Notice that

the conditional correlation between crude oil futures and soybean futures are positive and mostly larger than 0.5. This implies that portfolio investors have little scope for diversifying their portfolios between these two series. The conditional correlation between: crude oil futures and maize futures; crude oil futures and wheat futures; maize futures and soybean futures; maize futures and wheat futures and soybean futures and wheat futures do vary in sign and ranges between: -0.50 to 0.75; -0.75 to 0.50; -0.40 to 0.75; -0.65 to 0.75 and -0.75 to 0.60 respectively. The negative correlation enables the portfolio investors to diversify their price risk by including negatively correlated commodities in their portfolio. The conditional correlation between all pairs of futures prices reached low numbers around July-September, 2012 and July-September, 2014. This finding can be supported by the high food prices in the year 2012 which increased due to the Indian EXIM policy changes, financialization of Indian commodity market, U.S. drought in the year 2012 and Indian elections in the year 2014 followed by the increased supply of crude oil due to the decreasing energy demand from China and dwindling value of Indian rupee against U.S. dollar. The time-varying conditional correlation plots for spot market volatilities illustrate that the correlation for each pair of commodity is dynamic in nature and changes across time, indicating that traders and hedgers can manage the price risk of their portfolios by rebalancing it over time. The conditional correlation between crude oil spot and wheat spot and, maize spot and wheat spot touched the lowest value of -0.75 for the first time in July-September 2014. To the contrary, the correlation coefficient of crude oil spot and maize spot and, crude oil spot and soybean spot was positive and highest (+0.70 and +0.80 respectively) during the same time period. The conditional correlation dynamics differ for spot and futures markets. For instance, the correlation between crude oil spot and maize spot has changed from the later part of 2014 and reached -0.75. But, the conditional correlation graph of crude oil futures and maize futures does not show any such major movements in the correlation. Similarly, crude oil futures and soybean futures share positive relationship over the sample period while the conditional correlation between crude oil spot and soybean spot is not always positive. The time-varying conditional correlation results describe that the dynamics of correlations between all series changed significantly in July-September 2012 and July-September 2014 due to the reasons mentioned above.

As the results suggest that crude oil and food commodity prices have witnessed frequent fluctuations and there have been considerable cross-market volatility transmission. Using the conditional volatility estimates, we compute the hedge ratios for different pairs of commodities under study to suggest that how the price risks can be managed in these commodities (Kroner and Sultan, 1993). For a portfolio of two commodities i and j , a long position in one asset can be hedged with the short position in other asset to obtain the optimal hedge ratio. The optimal hedge ratio for the two assets in the portfolio can be computed as:

$$\theta_{ijt} = \frac{h_{ijt}}{h_{jtt}} \quad (3)$$

where θ_{ijt} is the optimal hedge ratio between the assets i and j , h_{ijt} represents the estimated conditional covariance between commodities i and j and h_{jjt} represents the conditional variance of the commodity j . Table 7 displays the average value of hedge ratios. The cheaper hedge consists of the lower hedge ratio and vice-versa. The graphs for most of the hedge ratios, estimated from the BEKK model, show significant variability during July-September 2012 and after July 2014 (see Fig. 4 and 5). The maximum value of many hedge ratios was recorded after July 2012 with the exception of crude oil spot and soybean spot and, wheat spot and maize spot which recorded the largest value of hedge ratios in the beginning of the sample period. The mean value of hedge ratio between crude oil futures and maize futures is 0.08 which means that a INR 1 long position in crude oil futures can be hedged by taking a short position in maize futures for INR 0.08. Overall, long wheat futures and short crude oil futures (0.01) and short soybean spot and long wheat spot (-0.28) are the cheapest hedge in spot and futures markets. The negative sign of the hedge ratio indicates that one should go short on the first asset and long on the other. The most costly hedge is long maize futures and short wheat futures (0.35) and, long crude oil spot and short soybean spot (0.30). The hedge ratios reported in Table 7 for spot and futures market exhibit that hedging in futures market is more costly compared to the spot market. The time-varying behavior of hedge ratios indicates that a dynamic hedging strategy is crucial for the efficient risk management.

The conditional volatilities from multivariate BEKK-GARCH model have been used to compute optimal portfolio weights to manage the risk of the investors (Kroner and Ng, 1998). The optimal portfolio weights can be computed as follows:

$$w_{ijt} = \frac{h_{j,j,t} - h_{i,j,t}}{h_{i,i,t} - 2h_{i,j,t} + h_{j,j,t}} \quad (4)$$

The optimal portfolio weight for commodity i would be:

$$w_{ijt}^* = \begin{cases} 0, & \text{if } w_{ijt} < 0 \\ w_{ijt} & \text{if } 0 \leq w_{ijt} \leq 1 \\ 1 & \text{if } w_{ijt} > 1 \end{cases}$$

where w_{ijt} is the weight of the first asset in a one rupee portfolio of two assets (i and j) at time t . h_{ijt} is the conditional covariance between the two assets, h_{jjt} is the conditional variance of asset j . The weight of the second asset is computed as $1 - w_{ijt}^*$. The summary of the portfolio weights estimated from BEKK model is tabulated in Table 8. The mean weight for CF/MF is 0.45, implying that for a INR 1 portfolio, 45 paise must be invested in crude oil futures and 55 paise invested in maize futures. The average weight for CS/MS is 0.23, indicating that for a INR 1 portfolio, 23 paise should be invested in crude oil spot and 77 paise invested in maize spot. The

portfolio weights for different combinations of selected commodities in futures market are more than the spot market.

Conclusion

The financial markets integration has transformed the nexus between crude oil and food commodity prices in non-food-input biofuel producing countries. Significant comovements are observed among selected commodities due to the increased financialization of commodities in recent years. In this paper, we examine level of interdependence and the short and long run dynamics of volatilities between crude oil and food commodity spot and futures prices. We propose the use of a multivariate BEKK-GARCH model to capture the short run and long run persistence in volatility between Indian crude oil, maize, soybean and wheat spot and futures prices for the period November 9, 2010 to December 31, 2015. We find the evidence of short term spillover from wheat futures to crude oil futures; maize futures to soybean and wheat futures; wheat futures to soybean futures. In the longer run, the findings show volatility spillover from crude oil futures to maize, soybean and wheat futures; and bidirectional volatility transmission between futures market of maize and wheat and, maize and soybean. Moreover, weak volatility persistence has been observed in the spot markets wherein the spillovers are from maize spot to wheat spot in short run and from maize spot to crude oil spot in the long run. Our findings indicate different structure of volatility spillover in the spot and futures markets of crude oil and food commodities. The cross-market volatility spillover is more evident in the futures market while own conditional volatility spillover is more significant in spot markets of these commodities. Overall, we find that volatility of crude oil futures spillovers to other commodity futures in the long run while in short run, it reacts to wheat futures. Moreover, the empirical findings indicate no volatility transmission between the spot markets of crude oil and food commodities in the short run. But, crude oil spot reacts to the volatility in maize spot market in the long run.

Our results have important implications for participants of spot and futures markets. We find the evidence of weak volatility dynamics in the spot market, wherein volatility of wheat and crude oil reacts to volatility of maize in the short run and long run respectively. This is a valuable finding for traders trading in wheat and crude oil markets as they can predict these prices based on the fluctuations in maize prices. Furthermore, futures market has shown more cross-market volatility persistence in the long run. Volatility of selected food futures is reacting to volatility of crude oil futures in the long run, indicating that portfolio investors should not buy and hold all these commodities in a single portfolio for a longer period. Our findings provide strong evidence that concerns about crude oil being the reason of volatile agricultural commodity prices are justified even in the non-food-input biofuel producing countries.

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Appendix

Table 1. Descriptive Statistics of Daily Returns

	DCF	DCS	DMF	DMS	DSF	DSS	DWF	DWS
Mean	-0.0411	-0.0430	0.0385	0.0541	0.0447	0.0486	0.0265	0.0274
Median	0.0177	0.0000	0.0000	0.0000	0.0519	0.1131	0.0000	0.0000
Maximum	9.9789	12.9608	13.5662	7.8781	8.7869	6.7876	7.9017	7.3236
Minimum	-17.4907	-17.7876	-10.5927	-12.2455	-29.6038	-21.7796	-15.1981	-5.6446
Std. Dev.	0.0191	0.0220	0.0162	0.0104	0.0191	0.0150	0.0119	0.0068
Skewness	-0.7410	-0.7860	0.6994	-1.1488	-5.9952	-5.2233	-1.7503	1.1583
Kurtosis	9.8295	8.8299	12.4112	27.5773	88.9279	65.0005	31.8758	27.0243
J-B	4405.5399	3586.2178	6954.7485	34141.3814	358982.7420	193232.8465	45845.9956	32798.9676
Prob	0	0	0	0	0	0	0	0
Obs	1070	1070	1070	1070	1070	1070	1070	1070

Note: Descriptive statistics is based on the returns. DCF- crude oil futures returns; DCS- crude oil spot returns; DMF- maize futures returns; DMS- maize spot returns; DSF- soybean futures returns; DSS- soybean spot returns; DWF- wheat futures returns; DWS- wheat spot returns.

Table 2. Correlation Matrix (Daily Returns)

Futures				
	DCF	DMF	DSF	DWF
DCF	1	0.0738	0.2666	0.0344
DMF	0.0738	1	0.2700	0.1864
DSF	0.2666	0.2700	1	0.0597
DWF	0.0344	0.1864	0.0597	1
Spot				
	DCS	DMS	DSS	DWS
DCS	1	-0.1076	0.2240	0.0279
DMS	-0.1076	1	0.0238	0.0241
DSS	0.2240	0.0238	1	-0.0387
DWS	0.0279	0.0241	-0.0387	1

Table 3. Correlation Matrix (Squared daily Returns)

Futures				
	DLCF	DLMF	DLSF	DLWF
DLCF	1	0.2844	0.3201	0.0330
DLMF	0.2844	1	0.2443	0.1953
DLSF	0.3201	0.2443	1	0.1371
DLWF	0.0330	0.1953	0.1371	1
Spot				
	DLCS	DLMS	DLSS	DLWS
DLCS	1	0.3981	0.1750	0.0898
DLMS	0.3981	1	0.1811	0.2325
DLSS	0.1750	0.1811	1	0.1713
DLWS	0.0898	0.2325	0.1713	1

Table 4. Unit Root Test results				
	ADF Test		PP Test	
	Level	Level 1	Level	Level 1
Crude oil spot	-0.47	-36.798*	-0.614	-36.606*
Crude oil futures	-0.514	-31.872*	-0.642	-31.914*
Maize spot	-1.971	-12.190*	-1.85	-33.537*
Maize futures	-2.478	-32.011*	-2.727	-32.076*
Soybean spot	-2.03	-11.254*	-1.933	-29.389*
Soybean futures	-1.804	-31.126*	-1.931	-31.187*
Wheat spot	-0.837	-26.886*	-0.847	-26.894*
Wheat futures	-1.339	-25.389*	-1.353	-32.130*

*significant at 1%

Table 5. BEKK-MGARCH Parameter Estimates

Variables	Futures Prices			Spot Prices		
	Coeff	T-Stat	Signif	Coeff	T-Stat	Signif
Mean equation						
m ₁₀	-0.0104	-0.1884	0.8506	0.0014	0.0217	0.9827
m ₁₁	0.0194	0.4873	0.6260	-0.1308	-3.8950	0.0001
m ₁₂	-0.0163	-0.4831	0.6291	0.0593	0.5833	0.5597
m ₁₃	0.0061	0.2327	0.8160	0.1120	2.5745	0.0100
m ₁₄	-0.0562	-1.3313	0.1831	0.0554	0.4038	0.6863
m ₂₀	0.0306	0.5750	0.5653	0.0160	0.5348	0.5928
m ₂₁	0.0058	0.2546	0.7991	0.0010	0.0599	0.9522
m ₂₂	0.0415	1.1459	0.2518	0.1711	0.9958	0.3194
m ₂₃	0.0364	0.8012	0.4230	0.0169	0.3499	0.7264
m ₂₄	-0.0209	-0.2605	0.7945	0.0627	0.5511	0.5815
m ₃₀	0.0555	1.0493	0.2940	0.1354	2.9036	0.0037
m ₃₁	-0.0007	-0.0223	0.9822	-0.0336	-1.8139	0.0697
m ₃₂	-0.0117	-0.2841	0.7764	0.0647	1.4855	0.1374
m ₃₃	0.0618	1.4363	0.1509	0.1289	1.6906	0.0909
m ₃₄	-0.0070	-0.1397	0.8889	0.3124	2.7150	0.0066
m ₄₀	0.0303	0.6646	0.5063	-0.0012	-0.0760	0.9394
m ₄₁	0.0585	1.9967	0.0459	0.0023	0.2356	0.8137
m ₄₂	-0.0282	-0.9325	0.3511	0.0318	0.2437	0.8074
m ₄₃	0.0045	0.1585	0.8741	-0.0046	-0.2020	0.8399
m ₄₄	-0.0276	-0.5632	0.5733	0.1430	0.7966	0.4257
Variance equation						
c ₁₁	0.1017	0.8243	0.4097	0.2303	1.4141	0.1573
c ₂₁	-0.1853	-0.1999	0.8416	0.3810	1.8441	0.0652
c ₂₂	0.8781	3.7700	0.0002	0.1798	1.2710	0.2037
c ₃₁	-0.2353	-0.5300	0.5961	-0.4318	-2.3002	0.0214
c ₃₂	-0.5479	-2.1260	0.0335	0.0418	0.1590	0.8736
c ₃₃	0.0000	-0.0002	0.9999	0.1949	0.8918	0.3725
c ₄₁	-0.2582	-0.6643	0.5065	0.0109	0.0584	0.9534
c ₄₂	-0.3564	-1.4605	0.1441	0.1194	1.4870	0.1370
c ₄₃	0.0000	0.0001	0.9999	-0.0600	-0.6353	0.5252
c ₄₄	0.0000	0.0000	1.0000	0.0000	0.0004	0.9997
α ₁₁	0.0931	2.3062	0.0211	0.1940	2.1141	0.0345

α_{12}	0.0302	0.4188	0.6753	0.0688	0.4900	0.6241
α_{13}	0.1150	1.0950	0.2735	-0.0701	-0.5828	0.5600
α_{14}	0.1928	3.2567	0.0011	-0.0086	-0.3755	0.7073
α_{21}	-0.0177	-0.3806	0.7035	0.0915	0.8396	0.4011
α_{22}	0.3810	3.2196	0.0013	0.4785	4.1120	0.0000
α_{23}	0.0112	0.0568	0.9547	0.0249	0.0851	0.9321
α_{24}	0.0227	0.3750	0.7077	0.0998	0.9603	0.3369
α_{31}	-0.0233	-0.4261	0.6700	-0.0625	-0.9530	0.3406
α_{32}	0.4984	4.3502	0.0000	-0.1157	-0.7051	0.4807
α_{33}	0.0966	0.4858	0.6271	0.4340	3.5210	0.0004
α_{34}	0.1927	2.2659	0.0235	-0.0561	-0.5164	0.6056
α_{41}	0.0760	0.7169	0.4735	0.0257	0.2372	0.8125
α_{42}	-1.0369	-4.3637	0.0000	0.2058	2.0551	0.0399
α_{43}	0.0477	0.0709	0.9435	-0.0479	-0.1126	0.9104
α_{44}	-0.2161	-0.8665	0.3862	0.7224	6.1378	0.0000
β_{11}	0.9692	23.8596	0.0000	0.9707	32.2081	0.0000
β_{12}	-0.0559	-0.9503	0.3419	-0.0522	-2.0899	0.0366
β_{13}	-0.0533	-0.3986	0.6902	0.0545	0.9118	0.3619
β_{14}	0.0058	0.1334	0.8939	-0.0053	-0.2024	0.8396
β_{21}	0.2608	2.2177	0.0266	-0.0264	-0.4184	0.6757
β_{22}	0.2060	2.0857	0.0370	0.7810	7.7293	0.0000
β_{23}	0.8517	6.4627	0.0000	0.1835	1.1333	0.2571
β_{24}	-0.3324	-2.9004	0.0037	-0.0538	-0.4151	0.6781
β_{31}	-0.2293	-2.8085	0.0050	0.0336	0.7779	0.4366
β_{32}	-0.2365	-2.0326	0.0421	0.0564	1.6035	0.1088
β_{33}	-0.0609	-0.1838	0.8542	0.8386	6.7321	0.0000
β_{34}	0.0657	0.5706	0.5683	0.0068	0.1610	0.8721
β_{41}	0.2042	1.7900	0.0735	-0.0115	-0.2256	0.8215
β_{42}	0.6322	2.5087	0.0121	-0.0974	-1.1298	0.2585
β_{43}	0.4224	1.2665	0.2053	0.0491	0.4513	0.6517
β_{44}	0.7904	6.0332	0.0000	0.8038	17.9219	0.0000

Note- In the model developed with futures returns, variable order is CF (1), MF (2), SF (3), WF (4) and in the model developed with spot returns, variable order is CS (1), MS (2), SS (3) and WS (4). In mean equation, m_{12} denotes the effect of a one period lag MF returns on CF returns. In variance equation, c represents the constant terms, α represents the ARCH terms and β represents the GARCH terms. The coefficient α_{12} denotes the short run volatility spillover from maize to crude oil while β_{12} represents long-term volatility spillover from maize to crude oil.

Table 6. Univariate Diagnostics Tests for Standardized Residuals

	CF	MF	SF	WF	CS	MS	SS	WS
Q(20)r	19.623	26.226	15.747	27.681	25.732	79.173	46.766	30.172
p values	0.482	0.158	0.732	0.117	0.175	0.000	0.001	0.067
Q(20)r^2	4.345	24.317	7.134	17.807	8.027	35.164	13.019	14.861
p values	1.000	0.229	0.996	0.600	0.992	0.019	0.877	0.784

Note: CF- crude oil futures, MF- maize futures, SF- soybean futures, WF- wheat futures, CS- crude oil spot, MS- maize spot, SS- soybean spot, WS- wheat spot

Table 7. Hedge Ratios (Long/Short) Summary Statistics

Futures					Spot				
	Mean	StDev	Min	Max		Mean	St Dev	Min	Max
CF/MF	0.08	0.14	-0.92	1.18	CS/MS	-0.23	0.49	-1.99	1.64
CF/SF	0.24	0.10	-0.35	1.04	CS/SS	0.30	0.31	-1.06	1.61
CF/WF	0.08	0.27	-2.99	1.62	CS/WS	-0.19	0.84	-4.19	2.83
MF/CF	0.04	0.32	-6.79	1.43	MS/CS	-0.01	0.14	-0.68	1.12
MF/SF	0.19	0.32	-2.93	6.7	MS/SS	-0.05	0.25	-1.94	0.95
MF/WF	0.35	0.52	-4.08	3.89	MS/WS	0.09	0.65	-1.93	3.68
SF/CF	0.29	0.34	-1.1	3.47	SS/CS	0.13	0.23	-1.31	2.82
SF/MF	0.26	0.33	-1.47	2.44	SS/MS	-0.06	0.49	-3.17	3.4
SF/WF	0.17	0.58	-10.73	5.81	SS/WS	-0.28	0.91	-5.58	2.01
WF/CF	0.01	0.13	-2.32	0.85	WS/CS	-0.01	0.08	-0.55	0.66
WF/MF	0.14	0.16	-0.38	1.47	WS/MS	0.05	0.26	-0.78	2.15
WF/SF	0.13	0.18	-0.41	2.28	WS/SS	-0.03	0.18	-1.43	1.1

Table 8. Portfolio Weights Summary Statistics

Futures					Spot				
	Mean	St. Dev	Min	Max		Mean	St. Dev	Min	Max
CF/MF	0.45	0.18	0	1	CS/MS	0.23	0.12	0	1
CF/SF	0.49	0.20	0	1	CS/SS	0.29	0.20	0	1
CF/WF	0.29	0.12	0	0.98	CS/WS	0.12	0.14	0	0.96
MF/SF	0.55	0.20	0	1	MS/SS	0.64	0.14	0.04	1
MF/WF	0.30	0.15	0	1	MS/WS	0.28	0.23	0	1
SF/WF	0.30	0.12	0	1	SS/WS	0.20	0.17	0	1

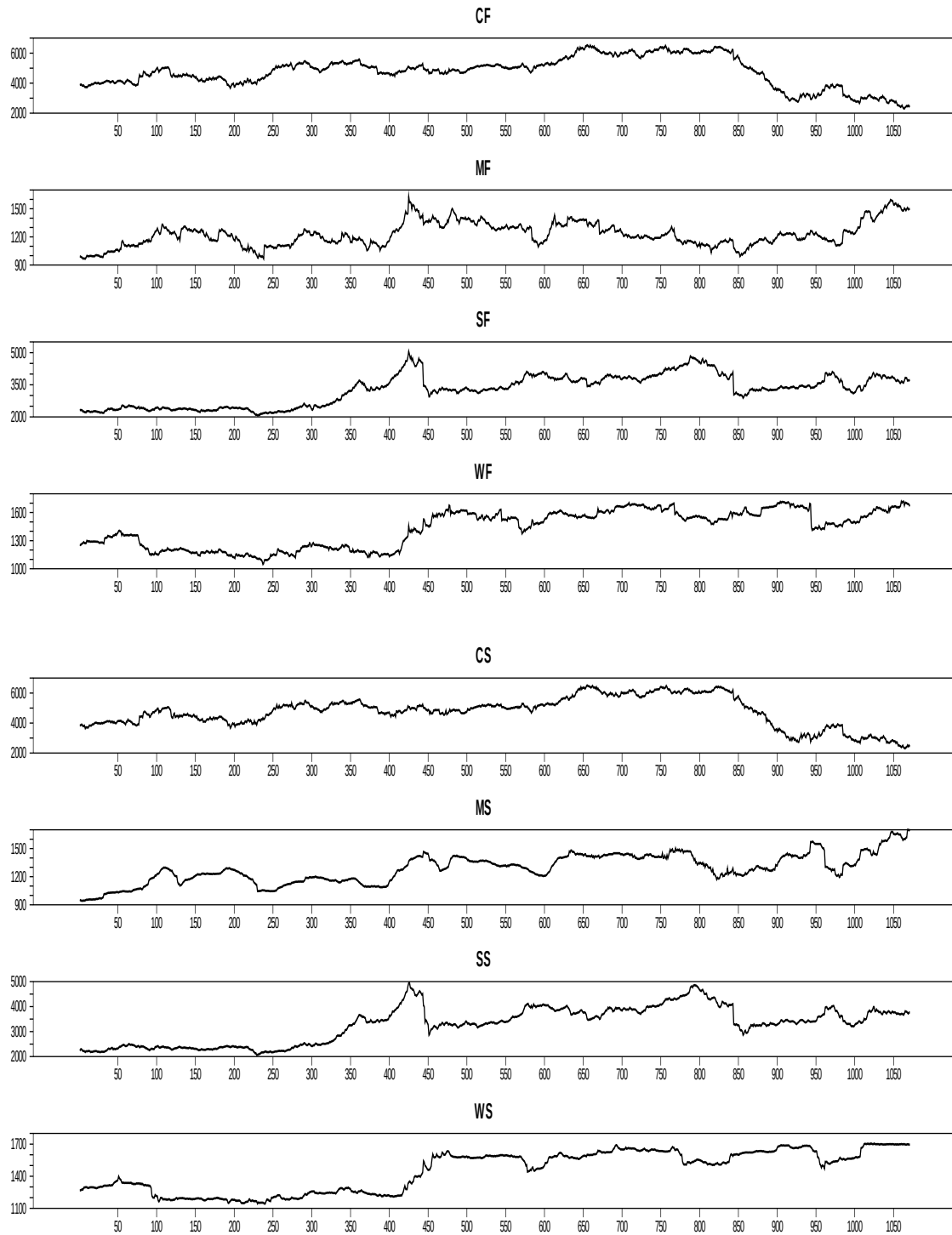


Fig.1. Time-series plot of spot and futures daily returns of crude oil, maize, soybean and wheat

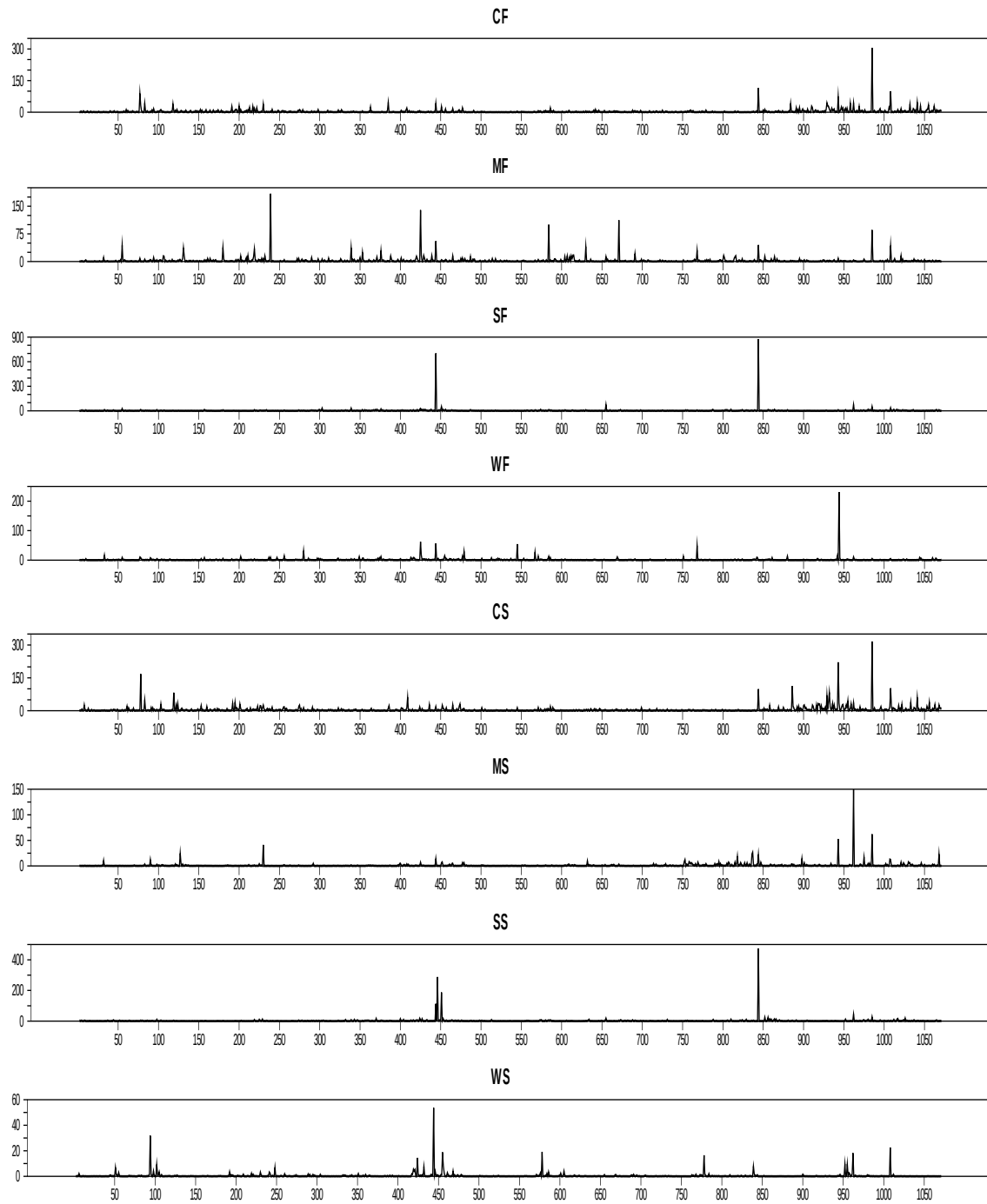


Fig.2. Time-series plot of spot and futures daily squared returns of crude oil, maize, soybean and wheat

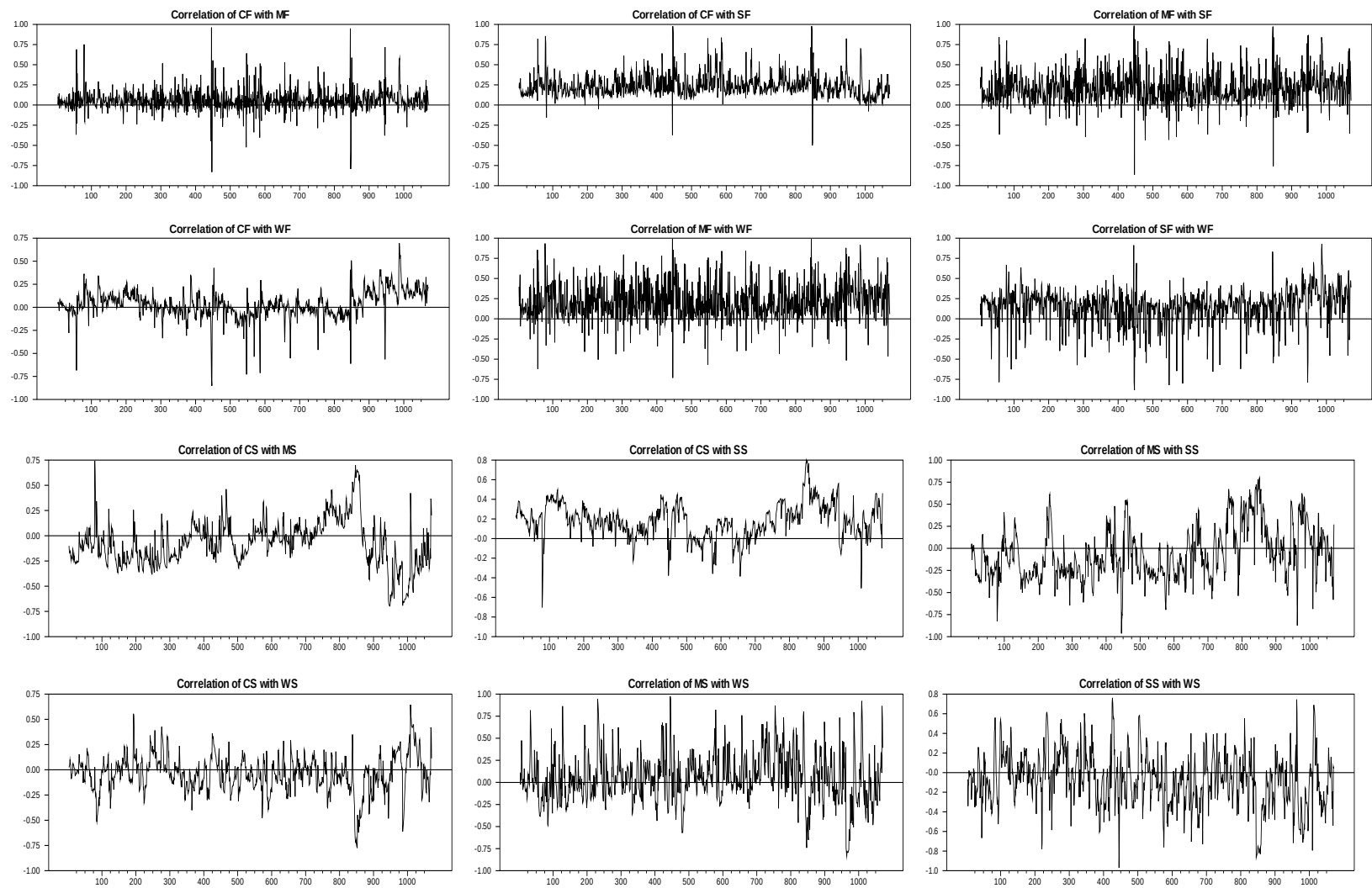


Fig.3. Time-varying conditional correlations from BEKK Model

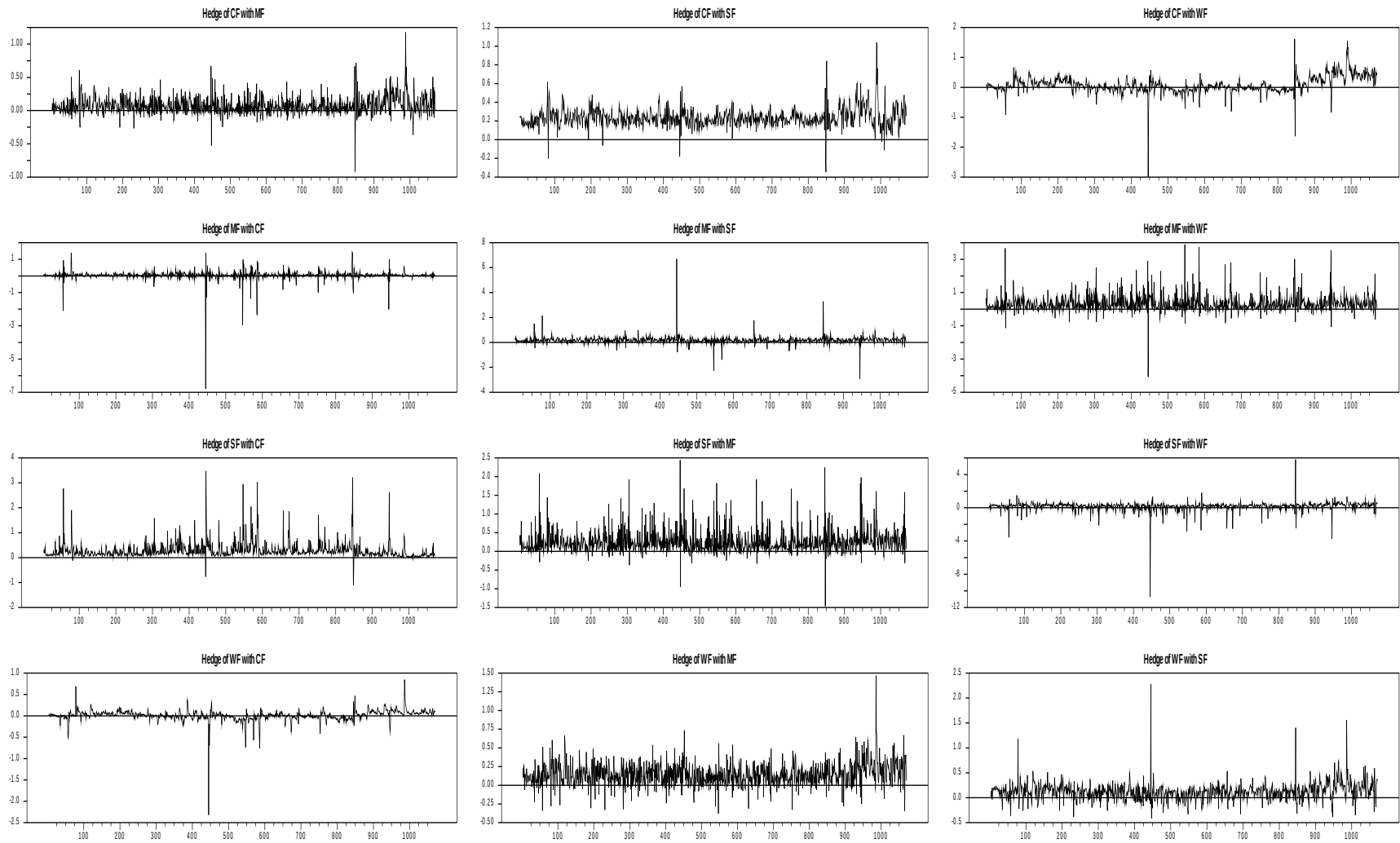


Fig.4. Time-varying hedge ratios for Futures market

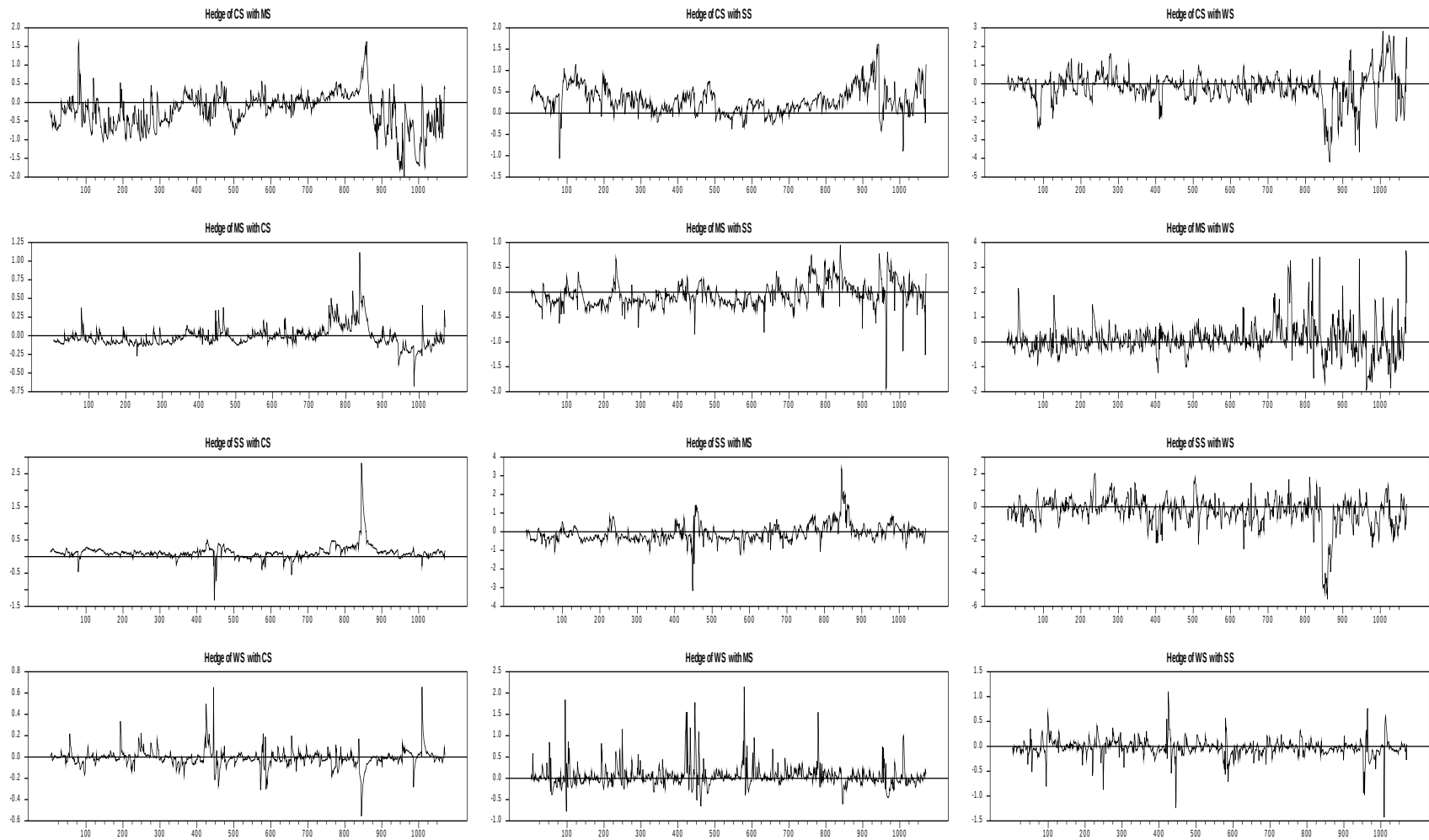


Fig.5. Time-varying hedge ratios for Spot market

